

Ans 1.
2025.09.10.

		A		B	
a	b	$a \leftrightarrow b$	$a \wedge b$	$(\neg a) \wedge \neg b$	$A \vee B$
i	i	i	i	f	i
i	f	f	f	f	f
f	i	f	f	f	f
f	f	i	f	t	t

a	b	$a \rightarrow b$	$\neg a \vee b$
i	i	i	t
i	f	f	f
f	i	i	t
f	f	i	t

$$\begin{cases} (a \wedge b \wedge c) \rightarrow d = \neg(a \wedge b \wedge c) \vee d = \neg a \vee \neg b \vee \neg c \vee d \\ a \rightarrow (b \rightarrow (c \rightarrow a)) = \neg a \vee (\neg b \vee (\neg c \vee a)) \end{cases}$$

$$\neg((a \wedge b) \vee c) \rightarrow a \leftrightarrow b = * \quad (a \vee b) \wedge c = (a \wedge c) \vee (b \wedge c)$$

$$(a \vee b \vee \neg c) \wedge (a \vee \neg b \vee c) \wedge (\neg a \wedge \neg b)$$

~~$$\begin{aligned} * &= \neg((a \wedge b) \vee c) \leftrightarrow b = ((a \wedge b) \vee c \wedge \neg a \wedge \neg b) \vee \\ &\vee ((\neg(a \wedge b) \wedge \neg c \vee a) \wedge \neg b) = ((a \wedge b) \vee c \wedge \neg a \wedge \neg b) \vee \\ &\vee ((\neg a \wedge \neg b) \wedge \neg c \vee a) \wedge \neg b = (a \vee c) \vee (b \wedge c) \wedge \neg a \wedge \neg b \vee \\ &\vee ((\neg a \wedge \neg c) \vee (b \wedge \neg c) \vee a) \end{aligned}$$~~

a	b	c	A	B
i	i	i	h	
h	i	i	i	
i	h	i	i	
h	h	i	h	
i	i	h	h	
h	i	h	i	
i	h	h	i	
h	h	h	i	

Minden medve szereti a mézet.

M mézek hza B medvek hza
S szereti amit

$$\exists m \in M, \exists b \in B, \neg S(b, m)$$

Minden asszony életében $\exists$ pillanatot, amikor olyat akar tenni,
amit nem szabad

$\varphi(b, k)$ boha megcsipte a huskét

$$\forall k \in K, \exists b \in B, \varphi(b, k)$$

$$\text{tag. } \exists k \in K, \forall b \in B, \neg \varphi(b, k)$$

$$\exists k \in K, \forall b \in B, \varphi(b, k) : \forall k \in K, \exists b \in B, \neg \varphi(b, k)$$

$$\forall b \in B, \exists k \in K, \varphi(b, k) : \exists b \in B, \forall k \in K, \neg \varphi(b, k)$$

$$(A \wedge B) \rightarrow (C \vee D) = (\neg A \vee \neg B) \vee C \vee D$$

$$(A \oplus B) \vee (C \wedge D) = \neg((A \wedge B) \vee (\neg A \wedge \neg B)) \vee (C \wedge D)$$

$$\neg((P \wedge Q) \vee (\neg P \wedge \neg Q)) \wedge (R \vee \neg(S \wedge T) \vee (\neg S \wedge \neg T)) \quad \boxed{\leftrightarrow}$$

$$(P \wedge \neg Q) \vee (\neg P \wedge Q) \wedge (R \vee (S \wedge \neg T) \vee (\neg S \wedge T)) \quad \boxed{\oplus}$$

$$\neg(P \Leftrightarrow Q) = P \oplus Q$$

2025.09.17.

gratulálok szerzett
egy tévő
+1 pont

$$\left(\bigcup_{i \in I} A_i \right) \cap \left(\bigcup_{j \in J} B_j \right) = \bigcup_{i \in I} \bigcup_{j \in J} (A_i \cap B_j)$$

$$\exists i \in I \exists j \in J : e \in A_i \wedge e \in B_j \quad \exists i \in I : e \in \bigcup_{j \in J} (A_i \cap B_j) \rightarrow$$

$$\rightarrow \exists i \in I \exists j \in J : e \in A_i \cap B_j$$

C egy alaphalmaz

$$C - \bigcup_{i \in I} A_i = \bigcap_{i \in I} (C - A_i)$$

$$e \in C \wedge \forall i \in I : e \notin A_i \quad \forall i \in I : e \in C \wedge e \notin A_i$$

$$x \in \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k \Leftrightarrow x \text{ véges sok kivétellel } \forall A_k - \text{ban benne van}$$

$$x \in \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k \Leftrightarrow x \text{ végtelen sok } A_k - \text{ban benne van}$$

$\Rightarrow$ kontrapozíció
 $\Leftarrow$

$$\exists i \in \{1, \dots, n\} \forall j \in \{k_1, \dots, n\} : x \in A_k$$

$$\forall k \in \{1, 2, \dots\} \exists j \in \{k, k+1, \dots\} : x \in A_j$$

Δ hsz $H = \{0, 1\}$ $p: H \times H \rightarrow H$ $p(a, b) = p(b, a)$ $A \subset X$

$$A' = \{x \in X \mid \forall a \in A: p(x, a) = 1\}$$

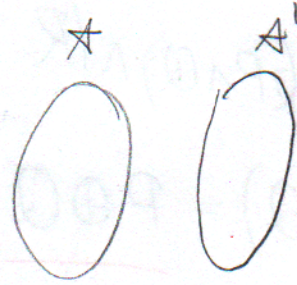
a) $\forall A, B \subset X \quad A \subset B \rightarrow B' \subset A'$

b) $\forall A \subset X \quad A \subset A''$

c) $A' = A'''$

d) $H = \mathbb{R}$

e) $H = \{1\}$



e) $X' = X$

a) $(x \in B \rightarrow x \in A) \rightarrow (\neg \forall a \in A, p(x, a) = 1)$

b) A' azon elemek, melyekhez p 1-et rendel

$$p(a', a) = 1$$

$$p(x, a'') = 1$$

a) $A = U(P(A))$

a2) $UB \subset A \leftrightarrow B \subset P(A) \xrightarrow{f} B \in P(P(A))$

b) $\{x, y\} \in A \rightarrow x, y \in U A$

d) f fo $\text{Ran}(f), \text{Dom}(f) \in P(U \cup f)$

c) $(x, y) \in A \rightarrow x, y \in U \cup A$

e) $f: A \rightarrow B$

$f \in P(P(P(A \cup B)))$

~~d) f fo. $\text{Ran}(f), \text{Dom}(f)$~~

$$1) A \text{ hz } \rightarrow \{A\}$$

$$2) A, B \text{ hz } \rightarrow \{A, B\}$$

$$3) A, B, C \text{ hz } \rightarrow \{A, B, C\}$$

$$c) (A \wedge \neg B) \vee (\neg A \wedge \neg C) \vee (B \wedge C) \vee (A \oplus C)$$

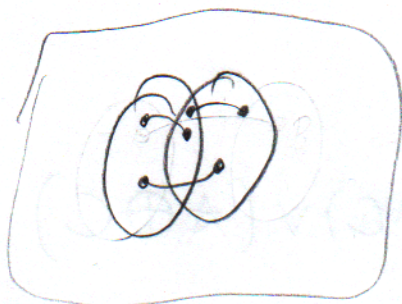
A	B	C	$A \wedge \neg B$	$\neg A \wedge \neg C$	$B \wedge C$	$A \oplus C$	
1	1	1	0	1	1	1	1
1	1	0	0	0	1	0	1
1	0	1	1	1	0	1	1
1	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	0	0	0	0	1	1
0	0	1	0	0	0	0	0
0	0	0	0	0	0	1	1

$(C \vee A)$	$(C \vee B)$	
1	1	
1	1	
1	1	
1	0	
1	1	
1	0	
1	0	
0	1	
0	0	

$$A \setminus B \subset X$$

2025.09.24.

$$R[A] \in B \Leftrightarrow R^{-1}[X-B] \subset X-A$$



$$R^{-1}[X-B] \subsetneq X-A \rightarrow$$

$$\exists x \in R^{-1}[X-B] \wedge x \notin X-A \Leftrightarrow x \in A \rightarrow x \notin R$$

$\Rightarrow$

$$\exists a \in R^{-1}[X]$$

$$\mathcal{H} \subset \mathcal{P}(A, B) \quad f: A^2 \rightarrow B$$

$$h_1, h_2 \in \mathcal{H}, h_1 \subset h_2 \vee h_2 \subset h_1$$

$$\bigcup \mathcal{H} \in \mathcal{H}$$

$$\exists i \in I, h_i \in \mathcal{H}, x \in h_i$$

$$a) f = \bigcup \mathcal{H} \text{ függvény}$$

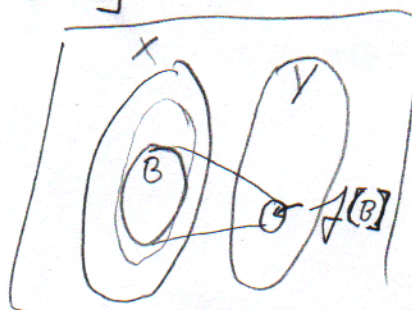
$$(a, c), (a, b) \in f \rightarrow \exists h_i, h_j \in \mathcal{H}: (a, c) \in h_i, (a, b) \in h_j \rightarrow$$

$$(h_i \subset h_j \rightarrow b \neq c) \vee (h_j \subset h_i \rightarrow b \neq c)$$

b) nézzük $(b, a), (c, a)$ -ra ugyanaz

$$f: X^2 \rightarrow Y \quad A \subset X \rightarrow A \subset f^{-1}[f[A]]$$

$$B \subset Y \rightarrow f[f^{-1}[B]] \subset B$$



f gürjektiv, akkor

$$x_n \leq y_n \Leftrightarrow (x_n - y_n \rightarrow 0) \vee (x_n < y_n \text{ } \forall n \text{ re v\u00e9ges sok h\u00edv\u00e9telleb})$$

i) $\trianglelefteq$ reflex\u00edv

ii) nem

iii) nem

iiii) nem

$\mathbb{Q}$

$\mathbb{R}$

$\mathbb{Z}$

nem Cauchy'
 $(-2)^n \quad (-2)^{n+1}$

iv)

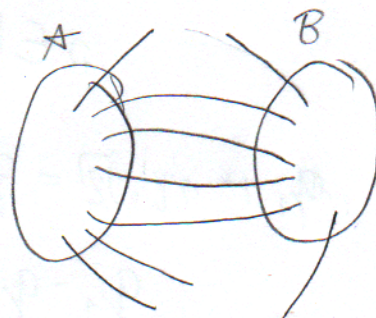
$$\left((-1)^n + \frac{1}{n} \right) \quad \left((-1)^n + \frac{1}{n^2} \right)$$

$$1 + \frac{2}{n^2}$$

~~ANZ~~
~~Adat~~

2025.10.01.

$$R[A] \subset B \Leftrightarrow R^{-1}[B^c] \subset A^c$$



(4.1.) $[0, 1] \rightarrow [0, 1]$

eg id.

$$f: x \mapsto x \text{ ha } x \in [0, 1] - \mathbb{Q}$$

$r_0, r_1, r_2, \dots$ a racional\u00edsok

$$r_0 \rightarrow r_1 \rightarrow r_2 \rightarrow r_3$$

teh\u00e1t $r_k \mapsto r_{k+1}$ teh\u00e1t „arr\u00e9ltoljuk”

d) $[2, 5] \rightarrow [10, 20]$

id helyett $\frac{10}{3}x + \frac{10}{3}$, felsoroljuk a $\mathbb{Q}$ -kat \u00e9s a 10-et \u00e9s 20-at

$$\text{ha } Q \text{ belüli } r_k \mapsto \frac{10}{3} r_{k+1} + \frac{10}{3}$$

$$\text{vagy }]-1, 1[\rightarrow [-1, 1]$$

$$f(x) = \begin{cases} -x - \frac{3}{2} & x \in]0, -\frac{1}{2}] \\ -x + \frac{3}{2} & x \in [\frac{1}{2}, 1[\\ 0 & \end{cases}$$

valami Cantor halmaza-
gosan

(nem tudom eldögni)

$$R(f) = [-1, -\frac{1}{2}] \cup \{0\} \cup]-\frac{1}{2}, 1]$$

$$e) f: \mathbb{R} - \mathbb{Q} \xrightarrow{\varphi} \mathbb{R}$$

$$Q \subset H = \left\{ \underbrace{0}_{\in \mathbb{Q}} = q + n\sqrt{2} \mid n \in \mathbb{N}_0 \right\}$$

$\mathbb{R}$ -ből $(\mathbb{R} - \mathbb{Q})$ -ba
vált ~~eg~~

$$R = H \cup R - H \quad f|_{R-H} = \text{id}$$

bijekció-e

$$0 \in H \quad (q + n\sqrt{2}) \mapsto q + (n+1)\sqrt{2}$$

$$q_1 + n\sqrt{2} = q_2 + m\sqrt{2}$$

$$q_1 - q_2 = (m - n)\sqrt{2}$$

(4.2.) trivi $\frac{1}{2}$ oldalú négyzetek

(4.3.) $1, a_1, a_2, a_3, \dots$

megoldottam, majd leírom, mert
táblában írtam

(b) $\left| \bigcup_{x \in \mathbb{R}} A_x \right| = |\mathbb{C}|$ de adjunkt meg bijektív

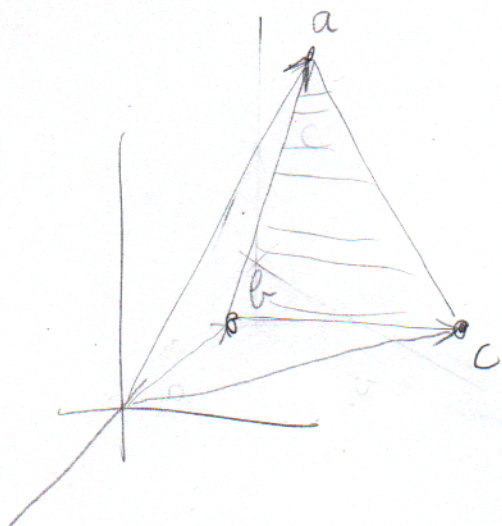
$\text{arctg}(x) \xrightarrow{\text{de}} \Delta$
 $\mathbb{R} \mapsto \text{számsz} \mapsto \text{szög}$

- (4.7) a) $2\mathbb{N}$ c) $\{2^k \mid k \in \mathbb{N}\}$
 b) $2\mathbb{N} + 1$ d) $\{p \in \mathbb{N} \mid p \text{ prim}\}$
 látható nem tudok rajzolni nincs pluszpozt.

(4.8) $(a_n, \dots, a_0)$ egész együtthatókból álló véges sorozat

(4.9) trivi

(4.10.)



π -hez nincs, sőt $\mathbb{R} - \mathbb{Q}$ -hoz sem, mivel:

$$T = \langle \underbrace{a-b}_{\in \mathbb{Q}}, \underbrace{c-b}_{\in \mathbb{Q}} \rangle \in \mathbb{Q}$$

(4.11.)

- a) $\underbrace{\kappa_0 + d}_{\text{az összes}}$ b) $\underbrace{C+d}_C$ c) $\underbrace{\kappa_0 d}_{C, \kappa_0}$ d) $\underbrace{C d}_C$

2025.10.08.

5.1

x^n $[0,1]$

hatarfuggo

$\forall b < 1$ -re $[0,b]$ $x^n \rightarrow x$

$|f_n - f| < b^n$

$|x^n - 0| < b^n \rightarrow 0$

eg a hatarfo

univerzalis
hugole

5.2.

a) $(\log x)^n$ $(\frac{1}{e}, e]$

hatarfo

b) $n \sin \frac{x}{n} = x \frac{\sin \frac{x}{n}}{\frac{x}{n}} \rightarrow x$ $\mathbb{R}$

c) $\frac{x^n}{1+x^{2n}} = |x| < 1 \rightarrow \frac{x^n}{1+x^{2n}} \rightarrow 0$

$\mathbb{R} - \{-1\}$

d) $\frac{2nx^4}{n^2x^4+n+3} = \frac{\frac{2}{n}x^4}{x^4 + \frac{1}{n} + \frac{3}{n^2}}$ $\mathbb{R}$

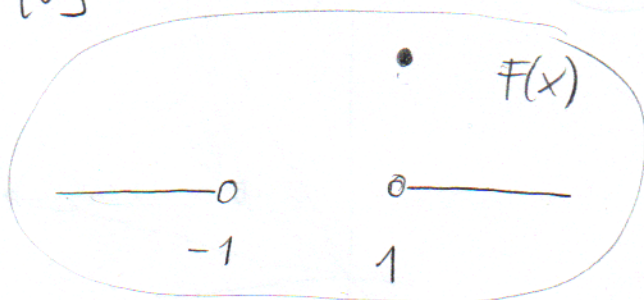
5.3

a) $f_n(x) = \frac{1}{x^n}$ $\text{Dom}(f_n) = \mathbb{R} - \{0\}$

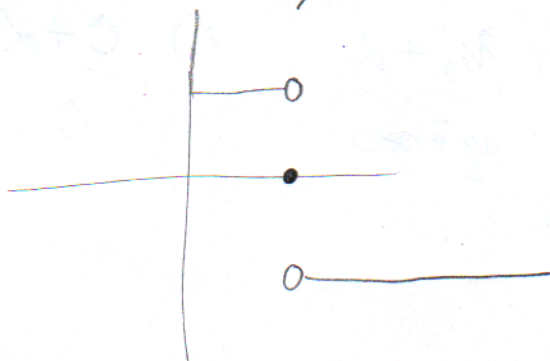
$|\frac{1}{x}| < 1 \rightarrow \frac{1}{x^n} \rightarrow 0$

$|\frac{1}{x}| > 1$

$x=1 \rightarrow \frac{1}{x^n} \rightarrow 1$



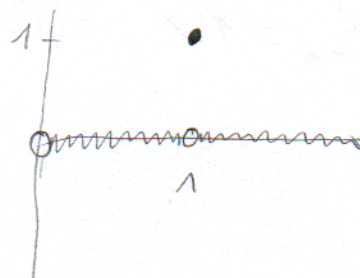
$\neq$



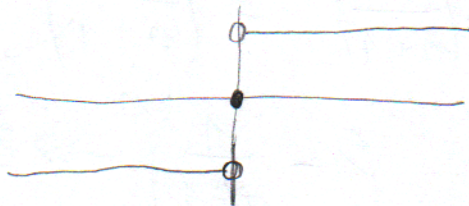
b) $\frac{1-x^{2n}}{1+x^{2n}}$ paros

c) $\frac{2x^{2n}}{1+x^{4n}}$ páros $|x| < 1$

$\Rightarrow$

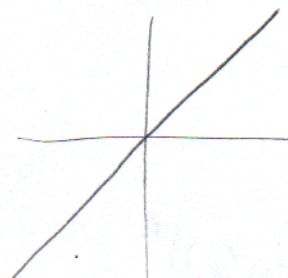


d) $\frac{e^{nx}-1}{e^{nx}+1}$



$\Rightarrow$

e) $n \log\left(1+\frac{x}{n}\right) = x$ $\mathcal{H}_H = \mathbb{R}$



$\Rightarrow$ mi a páráért nem?

f) $\frac{1}{(x+n)^2} \rightarrow 0$



$\Rightarrow$

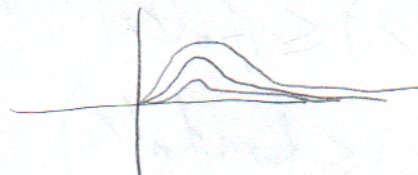
g) $\sum_{k=1}^n \frac{1}{k^x}$

$x > 1$ / $a > 1$ $[a, \infty)$

itt egyenletesen monoton egyfelé nem

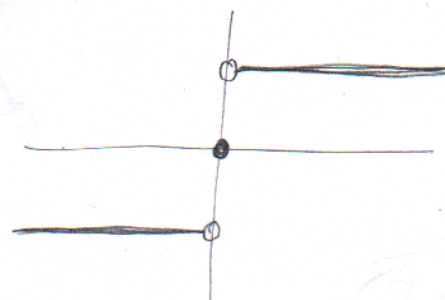
h) $x e^{-nx}$
mi?

$f'_n = e^{-nx} + x(-n)e^{-nx} = e^{-nx}(1-nx)$

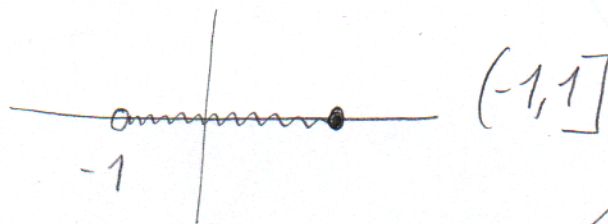


$f_n\left(\frac{1}{n}\right) = \frac{1}{n} e^{-1}$

i) $\frac{2}{\pi} \arctg(nx)$ $x > 0, f_n \rightarrow 1$
 $x=0$



j) $x^n - x^{n+1} = x(x^n - 1)$



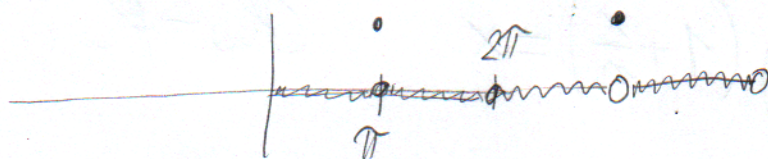
$$f'_n = nx^{n-1} - (n+1)x^n = x^{n-1}(n - (n+1)x)$$

$$\frac{n}{n+1} = x$$

$$\left(\frac{n}{n+1}\right)^n - \left(\frac{n}{n+1}\right)^{n+1} = \left(\frac{n}{n+1}\right)^n \left(1 - \frac{1}{n+1}\right)$$

$$\left(1 - \frac{1}{n+1}\right)^n \rightarrow \frac{1}{e} \quad |f_n - 0| < \frac{1}{n+1}$$

k) $(\sin(x))^n$



5.4 a) e^{-nx}

$$I_1 = [0, \infty) \quad I_2 = [a, \infty)$$

$$|e^{-nx} - 0| < e^{-na} = \left(\frac{1}{e^a}\right)^n \quad \text{itt venni kell vmi hűzőb-indexet}$$

5.7

$f_n: I \subset \mathbb{R} \rightarrow \mathbb{R}$ mon. növ. függvényekből álló

$\lim_{n \in \mathbb{N}} f_n(x) = f(x)$, ekkor $f(x)$ is monoton növekvő

$$f_n(x_1) \leq f_n(x_2) \quad \text{ha } x_1 < x_2$$

$$\lim f_n(x_1) \leq \lim f_n(x_2)$$

azaz a lim
monoton

$$f(x_1) \leq f(x_2)$$

5.8



$g: [0,1] \subset \mathbb{R} \rightarrow \mathbb{R}$ folyt. fv.

$$f_n: [0,1] \subset \mathbb{R} \rightarrow \mathbb{R} \quad f_n(x) = g(x)x^n \quad (n \in \mathbb{N}^+)$$

$g(1) \neq 0 \rightarrow f_n(1) \rightarrow g(1) \neq 0$ tehát van szakadási pont,

tehát nem $\Rightarrow$

$$g(1) = 0 \rightarrow |g(1)| < \varepsilon$$

$d(x)$

még nem írtam
csak annyit, hogy
 ε -hoz kell küszöbindexet
keresni és igenis
számít, hogy g felveszi
a maximumát.

5.9. $\lim_{n \rightarrow \infty} \int_2^5 nx e^{-nx} = \int_2^5 \lim_{n \rightarrow \infty} nx e^{-nx} = 0$

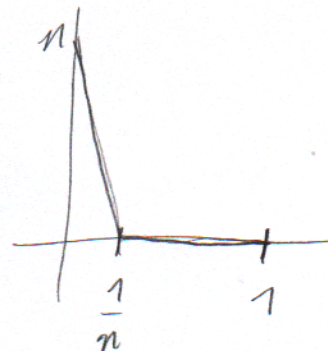
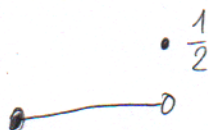
$$\left| x \frac{n}{(e^x)^n} - 0 \right| \leq \frac{5n}{2^{2n}}$$

$$\lim_{n \rightarrow \infty} \int_0^{2\pi} \frac{\arctg(n^5 x^2)}{x + \sqrt{n}} dx = 0$$

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{e^{nx}} = 0$$

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x} dx$$

$$g(x) = \frac{1}{1+x}$$



$$\left| \frac{\arctg(\quad)}{x + \sqrt{n}} \right| < \frac{\frac{\pi}{2}}{\sqrt{n}}$$

$$\left| \frac{x^n}{e^{nx}} - 0 \right| < \frac{1}{2^n} \quad x < \frac{1}{2}$$

$$< \frac{1}{\sqrt{e}^n} \quad x \geq \frac{1}{2}$$

$$\max(N_1, N_2)$$